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ACX7000 Hardware Profiles

An HPE Juniper Networking Techpost

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1 ACX7000 Hardware Profiles

Nicolas Fevrier - 01/24/2024

1.1 Introduction

The ACX7000 routers are powered by Broadcom Jericho2 Series chipsets. These Packet Forwarding Engines (PFEs) are equipped with a substantial internal memory pool known as Modular DataBase (MDB). During the system boot-up, the MDB is segmented into various specialized databases, each designated to store specific information types and accessible from distinct pipeline blocks.

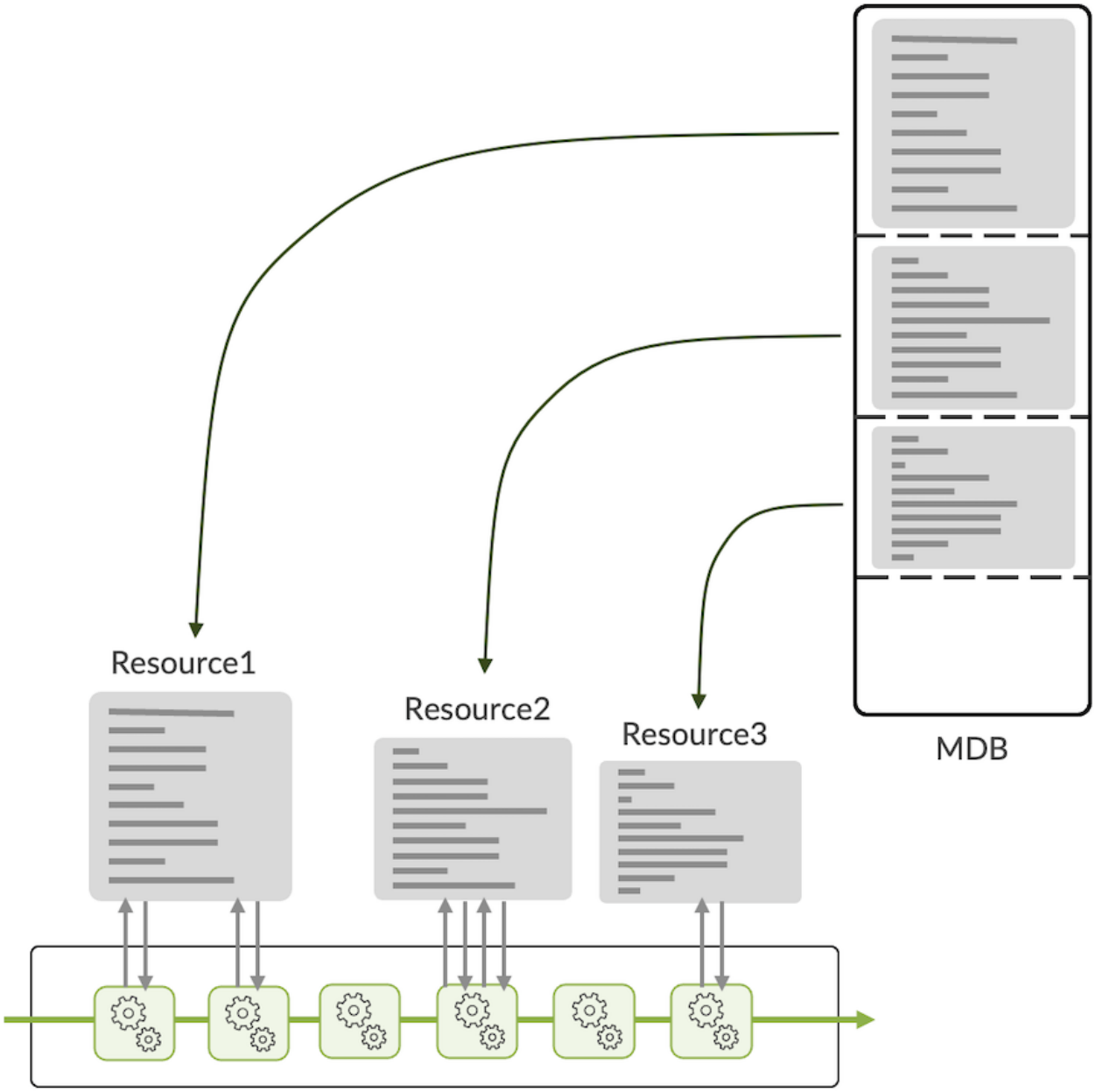


Figure 1: MDB, Resources and Pipeline

Depending on the way we allocate memory space for each database, the operator can decide to privilege one role over another. For example, it could be more L2-centric, or more L3-capable, or even more specialized in

subscriber-related functions. This allocation is pre-defined in what we call “hardware database profiles” (or simply “profiles”).

With the release of Junos 23.2 and then 23.4, several important improvements have been brought to the MDB management and the way routes are stored in the different memories. This article will elaborate on these enhancements and furnish the reader with the essential knowledge needed to determine the most suitable MDB profile for their requirements.

1.2 The Modular DataBase

The MDB is a memory present inside the Jericho2 forwarding ASIC. It’s meant to store operational information and not packets (it’s not a buffer).

Depending on the type of chipset, we will have two different sizes:

- 1.5M entries
- 3M entries

The following chart in diagram2 illustrates the MDB size present in each PFE and each ACX7000 router type.

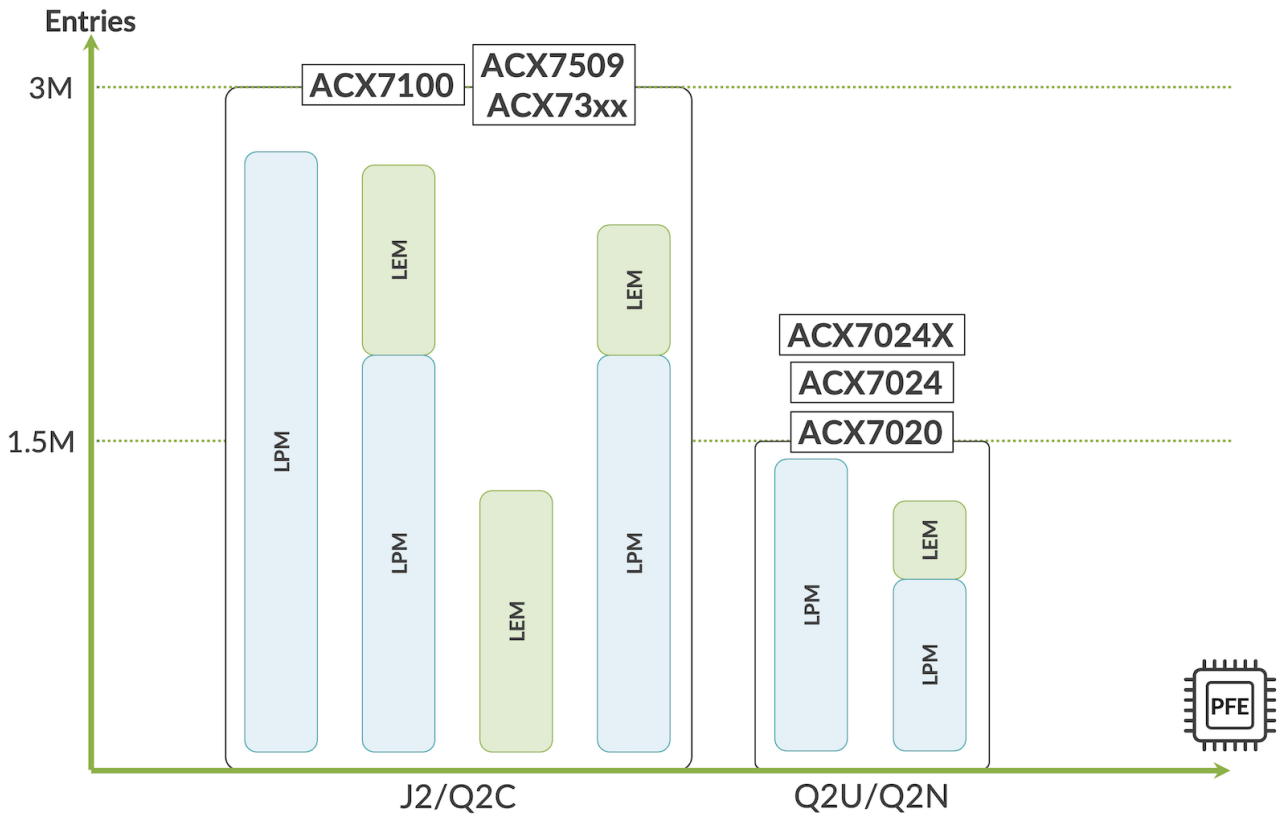


Figure 2: DNX Jericho2 PFE and Associated MDB Sizes

Operational information used by the different blocks in the pipeline is stored in these multiple databases. To name a few and not exhaustively:

- Interfaces (IFD/IFL)
- Routing Information
- Next-Hop

- Load-balancing information
- Encapsulation
- Tunnels
- Translation information
- VLAN manipulation information

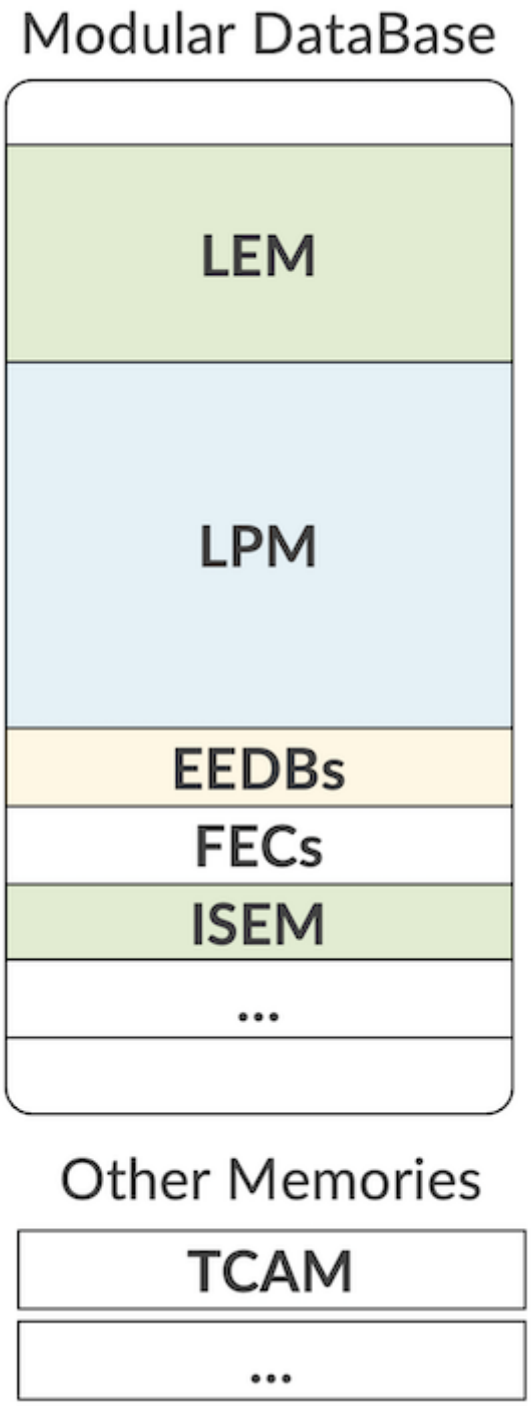


Figure 3: Illustration of Resources in MDB

First, let's try to clear a common misconception. Frequently, we receive questions about "the TCAM size" in this chipset. But what they actually meant is not TCAM but MDB resources and size. The MDB is an SRAM and not a TCAM.

It can create a confusion because we do have internal and potentially external TCAMs in ACX7000 routers. They are just different memory types serving different purposes:

- iTCAM (internal) is used to store firewall filters, BGP FlowSpec rules, and QoS/CoS information.
- eTCAM (external, OP2) can be used to store routes and filters but is only available on the ACX7332.

Other memories are used in the system that are not part of the MDB. For example, the Statistics (counters and meters).

Note: Diagram 1 illustrates that different ACX7000 routers come with different MBD sizes. It's important to keep in mind it'ss not the only parameter influencing the supported scale. For example, the ACX7024 (non-X) may hit memory (CPU DRAM) bottlenecks before reaching the MDB resource limits.

1.3 Where Do We Store the Routes?

In the most common scenarios, prefixes will be stored in two databases inside the MDB:

- The LPM: Longest Prefix Match
- The LEM: Large Exact Match

Like any other resources, the size of LEM and LPM will be directly influenced by the hardware profile enabled and how it carved the MDB at boot-up time.

Depending on the release running on the system, the behavior may be slightly different and the Junos release 23.2R1 marks a clear change.

1.3.1 Before Junos 23.2R1

The release 23.2R1 is the moment we changed the behavior. Before its introduction, we had host routes IPv4 and IPv6, MPLS labels, and MAC address stored in LEM, while all other prefixes, IPv4 and IPv6, were pushed in LPM.

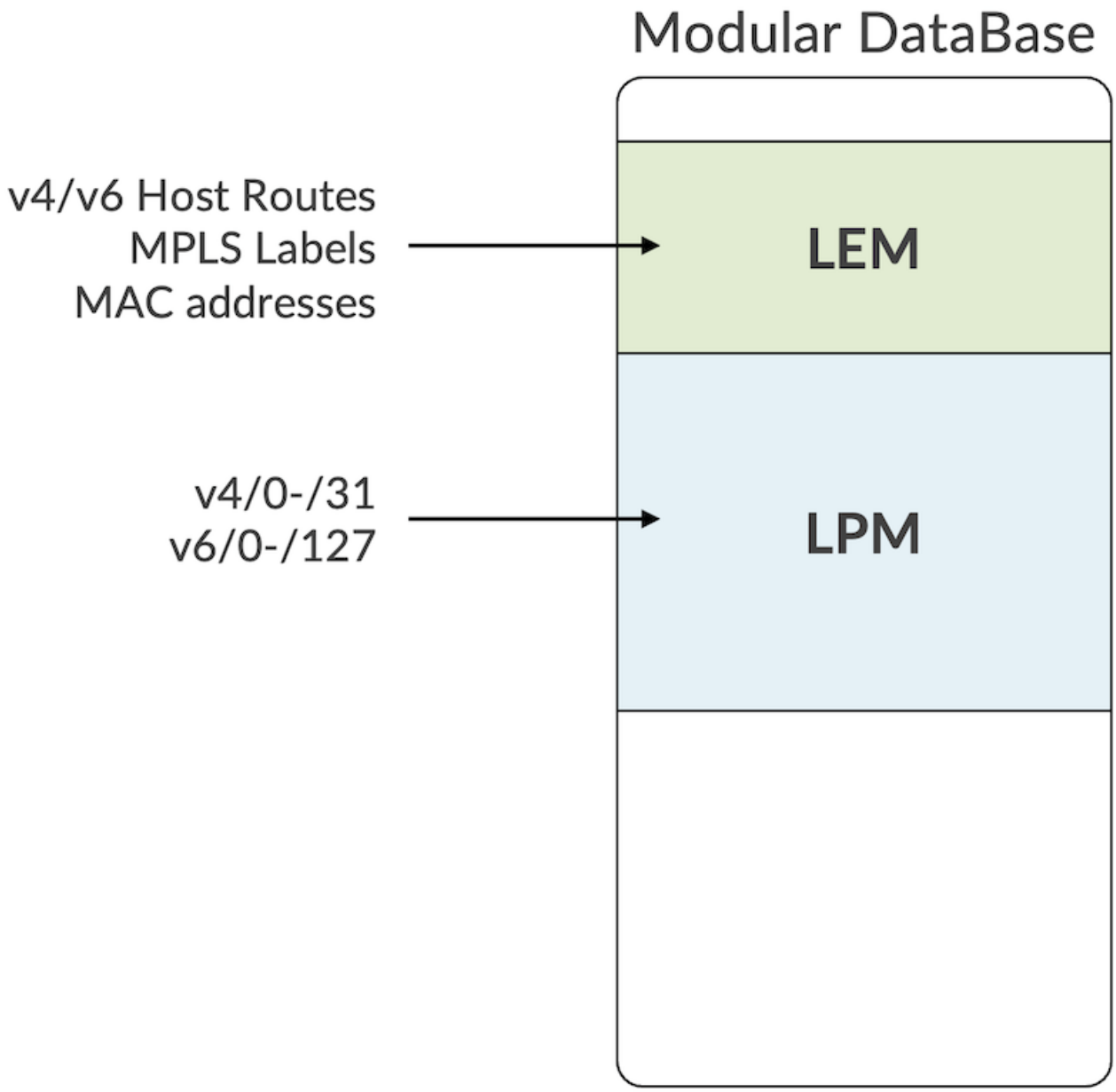


Figure 4: LEM and LPM roles before Junos 23.2R1

1.3.2 From Junos 23.2R1 Onwards

Starting from 23.2R1, we simplified the logic by moving all prefixes, host or not to LPM.

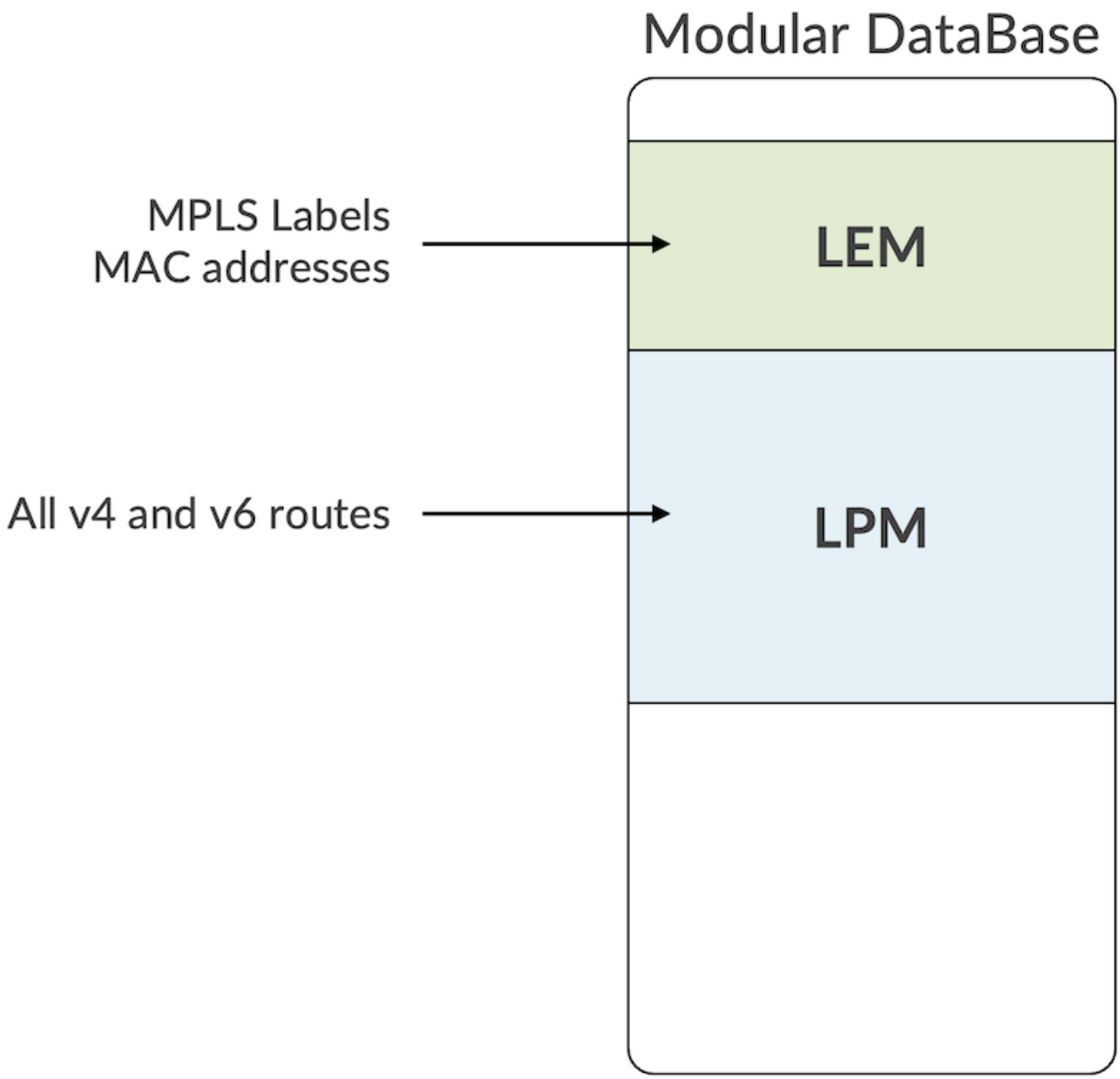


Figure 5: LEM and LPM roles starting from Junos 23.2R1

The main goal is: make things simpler. But also, it will eliminate the need to move prefixes from LEM to LPM when subsequent host routes are “aggregated” by the FIB compression algorithm. This cFIB feature is also introduced in Junos 23.2R1. You can read more on this topic here: https://techpost.hpe.com/articles/ptx-fib-compression/article

One important exception to this new behavior is when the carrier-ethernet profile is activated. We will detail these profiles below in the same article. But when carrier-ethernet is configured, the prefixes are stored with host v4/32 and v6/128 in LEM and the rest in LPM, like it was before 23.2R1.

1.4 MDB Profiles

Now that we explained where the routes are being stored, let’s study in detail this concept of MDB profiles and what recently changed.

1.4.1 Before Junos 23.3R1

Since we introduced the ACX7000 Series and just before the release of Junos 23.3R1, we used the following four profiles:

- balanced
- balanced-exem
- l2-xl
- l3-xl

With “balanced” being the default profile enable (if you don’t configure anything).

Also, a mechanism of “lpm-distribution” was needed to allocate specific space for routes in VRF (private) and routes in Global Routing Table (public).

So, prior to Junos 23.3R1, we had the following configuration options:

```
nfevrier@rtme-acx-48l-03# set system packet-forwarding-options hw-db-profile ?
Possible completions:
+ apply-groups Groups from which to inherit configuration data
+ apply-groups-except Don't inherit configuration data from these groups
  balanced Selects Balanced DB profile, restarts PFE
  balanced-exem Selects Balanced-Exem DB profile, restarts PFE
  l2-xl Selects L2-XL DB profile, restarts PFE
  l3-xl Selects L3-XL DB profile, restarts PFE
  lpm-distribution Specify route distribution between public and private(vrf/vpn) routes in lpm.
      Default is 1
[edit]

nfevrier@rtme-acx-48l-03# set system packet-forwarding-options hw-db-profile lpm-distribution ?
Possible completions:
  1 Set the lpm-distribution to 1
  2 Set the lpm-distribution to 2. Not valid for Balanced profile
  200 Set the lpm-distribution to 200. Valid only for Balanced profile
  3 Set the lpm-distribution to 3. Not valid for Balanced profile
  4 Set the lpm-distribution to 4. Valid only for l2-xl profile
[edit]

nfevrier@rtme-acx-48l-03#
```

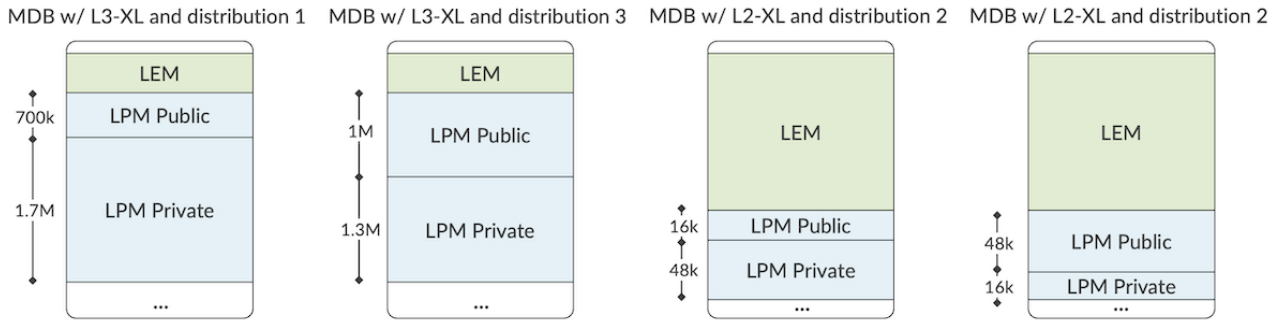


Figure 6: Examples of lpm-distribution

We can summarise the pre-23.3R1 with the following chart:

Default

Merged Domains

lpm-distribution	Domain	Balanced	Balanced-exem	L2-xl	L3-XL
1	Private	1,500,000	1,260,000	64,000	1,700,000
	Public		180,000	0	700,000
2	Private	N/A	1,100,000	48,000	1,500,000
	Public		370,000	16,000	900,000
3	Private		900,000	32,000	1,300,000
	Public		540,000	32,000	1,000,000
4	Private		N/A	16,000	N/A
	Public			48,000	
200	Private	1,500,000		N/A	
	Public	32,000			

Figure 7: Diagram 7: Chart Summarizing the different profiles and lpm-distribution options

While the balanced profile was merging the Private and Public domains from the get-go, things were a little bit more complicated for the other profiles, with these rpm-distributions. It led to weird situations where an L3-XL profile will offer less routing scale in the global routing table than the default Balanced profile.

It required simplification. And that's exactly what we did with Junos 23.3R1

1.4.2 From Junos 23.3R1 Onwards: New Profiles and New Default

The first modification comes with the renaming of the profiles to match "use-cases" and a new default profile.

Now, we have:

- **cloud-metro**
- **lean-edge** (the new default)
- **carrier-ethernet**
- **bng**

Former Profile	New Profile Name	
balanced	cloud-metro	
l3-xl	lean-edge	Default from JUNOS 23.3 onwards
l2-xl	carrier-ethernet	
balanced-exem	bng	

Figure 8: Mapping of old and new profiles

Today, most of the customers are using ACX7000 series with L3 scale preference. That's what motivates the change of default to lean-edge.

When you upgrade your router to the 23.3 release (or a later version), we have two situations.

- **1.** A profile was specifically configured before the upgrade. When the system reboots, it will enable the equivalent in the chart in diagram 7 above.
- **2.** If you don't have any profile configured before, the balanced profile is running on the router. After upgrading, the new default will be changed to lean-edge.

```
evo-pfemand[11984]: EVO_PFEMAND_MDB_PROFILE: Default MDB profile configured to "lean-edge"
```

Now you have these profile options in the config CLI for ACX7100, ACX7509, ACX7348, and ACX7024X:

```
nfevrier@rtme-acx-48l-03# set system packet-forwarding-options hw-db-profile ?
Possible completions:
+ apply-groups Groups from which to inherit configuration data
+ apply-groups-except Don't inherit configuration data from these groups
  bng Selects BNG DB profile, restarts PFE
  carrier-ethernet Selects Carrier Ethernet high scale MAC DB profile, restarts PFE
  cloud-metro Selects Cloud-Metro DB profile, restarts PFE
  lean-edge Selects Lean Edge max IPv4/IPv6 FIB DB profile, restarts PFE
[edit]
```

And for ACX7024, only cloud-metro and lean-edge

```
nfevrier@rtme-acx7024-08# set system packet-forwarding-options hw-db-profile ?
Possible completions:
+ apply-groups Groups from which to inherit configuration data
+ apply-groups-except Don't inherit configuration data from these groups
  cloud-metro Selects Cloud-Metro DB profile, restarts PFE
  lean-edge Selects Lean Edge max IPv4/IPv6 FIB DB profile, restarts PFE
[edit]
```

You'll note that lpm-distribution option is not present in the CLI and that's what we are going to explain in the next section.

1.4.3 From Junos 23.3R1 Onwards: Merge of Public and Private KAPS Domains

The lpm-distribution present in past versions for l2-xl, l3-xl, and balanced-exem required to know, in advance, exactly how many routes will be received in VRFs or in the inet.0/inet.6 tables. It was equivalent to sub-profiles, as illustrated in the diagram 6.

This approach was confusing for many operators since it was not present for the default balanced mode, and it lacked flexibility. In 23.3R1, we are simplifying this process by eliminating the entire concept of lpm-distribution: we are merging the public and private domains.

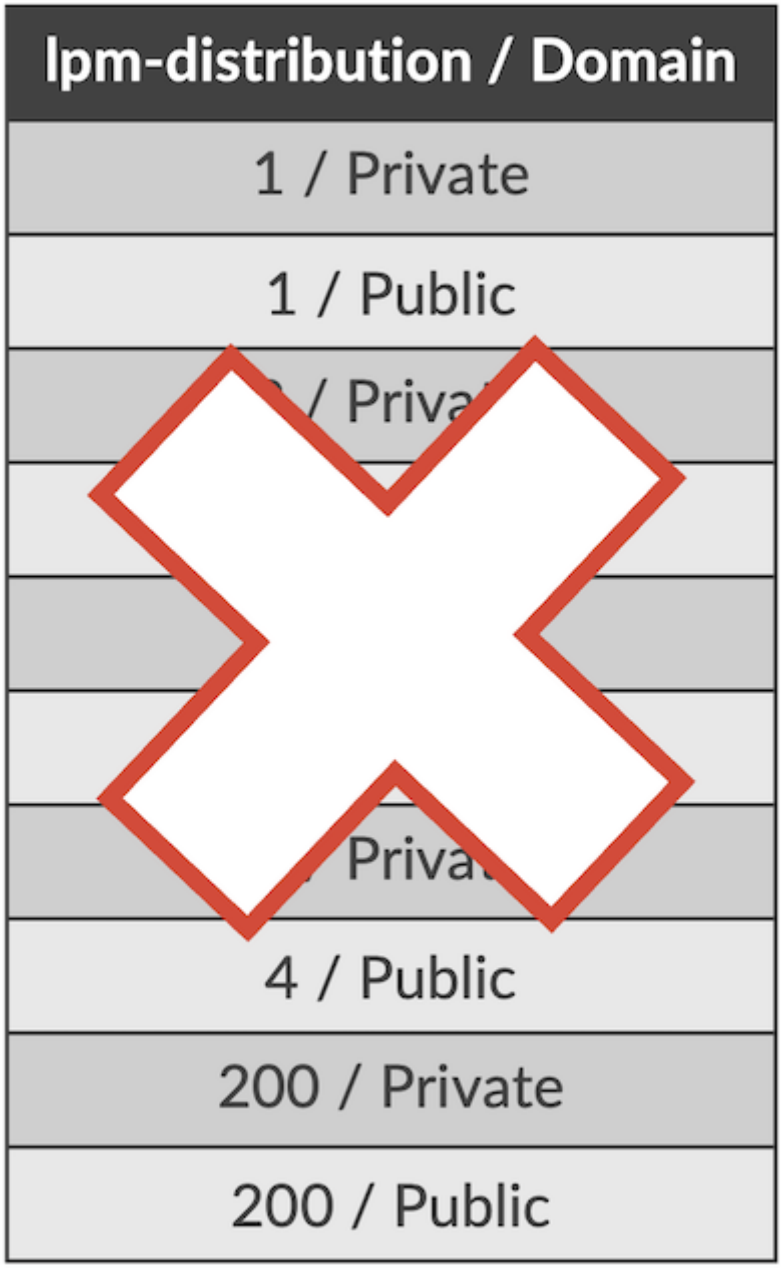


Figure 9: No more public and private domain differentiation.

That means, with 23.3R1, you don’t have to worry about the prefix length (host or not) or the domain, all the routes will go in LPM. If an lpm-distribution was configured in the previous version, it will be ignored when rebooting in 23.3R1 after the upgrade.

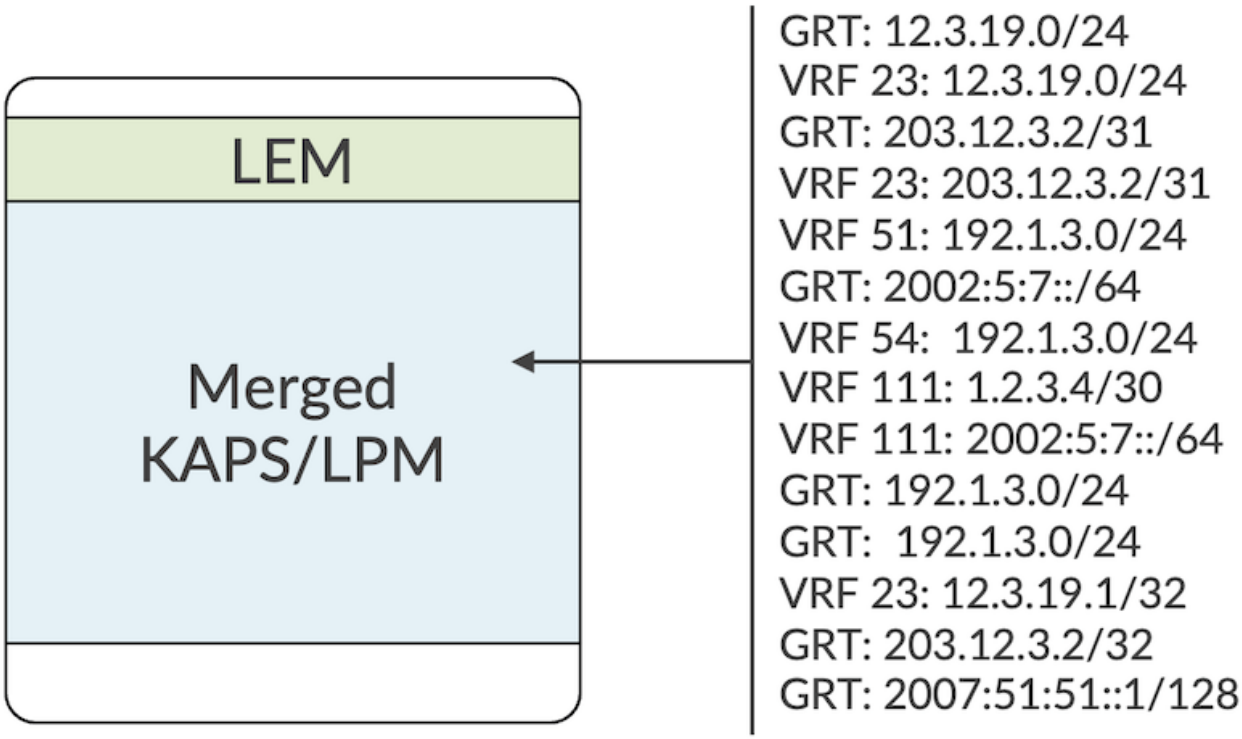


Figure 10: Simplification of the route-storing logic

To go a little bit deeper into the details, what I referred to as LPM above is the "KAPS1" size allocated from the system perspective. Another space, KAPS2, is allocated for (S,G) IPv6 multicast sources specifically: 32k sources occupying 96k entries for default and 1k source / 4k entries for carrier-ethernet profile.

It removes the limitation with IPv6 SSM that existed with some profiles in previous releases.

Platform		Lean-Edge (ex L3-XL)	Cloud-Metro (ex Balanced)	Carrier-Ethernet (ex L2-XL)	BNG (ex Balanced-exem)
ACX7100 ACX7509 ACX7348	LPM/KAPS1	2,273,280	1,515,520	61,440	1,326,080
	LPM/KAPS2	94,720	16,384	4,096	189,440
	LEM	158,921	731,045	1,017,109	508,552
ACX7024	LPM/KAPS1	1,136,640	663,040	N/A	N/A
	LPM/KAPS2	47,360	16,384	N/A	N/A
	LEM	393,216	1,245,184	N/A	N/A
ACX7024X	LPM/KAPS1	1,136,640	663,040	61,440	663,040
	LPM/KAPS2	47,360	16,384	4,096	94,720
	LEM	393,216	1,245,184	2,228,224	1,048,576

Default Mode

Figure 11: Profiles details per platform

For the most part, these improvements are not changing the features support and the performance of the ACX7000 Series. One exception to this statement: VRF fallback/route leaking. This feature was executed in one pass and now will require a second lookup now that we merged KAPS domains. The traffic will be recirculated (meaning, it will use the RCY interface).

A follow-up article will go deeper into the recycling interfaces, the bandwidth available for each platform and the features existing to monitor and manage them.

1.5 Resource Utilization

Let's focus with the utilization of LPM (KAPS1 and KAPS2) and LEM for each L2 and L3 information category.

Entries presented here are "IPv4 unicast route" equivalent and "-" means: "not used".

	LEM	LPM (KAPS1)	LPM (KAPS2)
IPv4 /0-/31	-	1	-
IPv4 /32	1 (with carrier-ethernet profile)	1 (with all other profiles)	-
IPv6 /0-/64	-	2	-
IPv6 /65-/127	-	2	-
IPv6/128	2 (with carrier-ethernet profile)	2 (with all other profiles)	-
MC (S,G)v4	-	2	-
MC (S,G)v6	-	3	3
MPLS	half entry (60b)	-	-
MAC Address	1	-	-

Notes:

- LEM can be store entries of 30bits, 60bits, 120bits and 240bits
- LPM physical DB is an algorithmic, compression-based database. As such, the number of entries KAPS can store (in a given MDB profile) is not deterministic but rather depends on the distribution of the key values.
- It's fair to consider that an IPv6 entry is using twice the size of an IPv4 entry in LPM.
- Also, FIB compression is another important factor that will come into play to reduce the occupied space. It's enabled by default on all ACX7000 products except the ACX7024.

1.6 Monitoring

To check the details of the active hardware profile, at the cli-pfe level we can use the following:

```
nfevrier@rtme-acx-48l-03:pfe> show evo-pfemand mdb-info
User Config : Profile:lean-edge Kaps:1
Converted Config: Profile:custom_lean_edge_jnpr Kaps:1
Max possible mac limit for this profile:155000
TableName Type Capacity EntrySize
================================================
NONE 0 0
TCAM 0 0
KAPS1 KAPS (LPM) 2273280 0
KAPS2 KAPS (LPM) 94720 0
ISEM1 Exact Match 98304 30
```

```
ISEM2 Exact Match 131072 30
ISEM3 Exact Match 131072 30
INLIF1 Direct Access 65536 60
INLIF2 Direct Access 65536 60
INLIF3 Direct Access 65536 60
IVSI Direct Access 43691 90
LEM Exact Match 655360 30
IOEM1 Exact Match 65536 30
IOEM2 Exact Match 98304 30
MAP Direct Access 0 0
FEC1 Direct Access 78644 150
FEC2 Direct Access 78644 150
FEC3 Direct Access 104858 150
PPMC Exact Match 131072 30
GLEM1 Exact Match 163840 30
GLEM2 Exact Match 196608 30
EEDB1 EEDB 49152 0
EEDB2 EEDB 49152 0
EEDB3 EEDB 65536 0
EEDB4 EEDB 65536 0
EEDB5 EEDB 32768 0
EEDB6 EEDB 32768 0
EEDB7 EEDB 32768 0
EEDB8 EEDB 32768 0
EOEM1 Exact Match 98304 30
EOEM2 Exact Match 65536 30
ESEM Exact Match 131072 30
EVSI Direct Access 65536 30
SEXEM1 Exact Match 131072 30
SEXEM2 Exact Match 0 0
SEXEM3 Exact Match 0 0
LEXEM Exact Match 262144 30
RMEP_EM Exact Match 98304 30
KBP 0 0

nfevrier@rtme-acx-48l-03:pfe>
```

And to check the resource utilization:

```
nfevrier@rtme-acx-48l-03:pfe> show evo-pfemand resource terse-usage
UNIT 0 MDB Profile: custom_cloud_metro_jnpr
Resource Usage Capacity size
COS_HR_ELEM 105 512 1
COS_VOQ 840 65536 1
COS_VOQ_CNCTR 840 98304 1
ECMP 32 32000 30
EEDB_1 156 49152 0
EEDB_2 36 49152 0
EEDB_3 5 65536 0
EEDB_6 30 32768 0
EEDB_7 163 0 0
EEDB_8 114 0 0
EGR_VSI 36 65536 30
FAILOVER 278 6000 30
FEC_1 68 52429 150
FEC_2 3 26215 150
FEC_3 111 52429 150
GLEM 21 64000 1
IFL_STATS 15 16000 1
ING_MC_GRP 9 262144 1
ING_VSI 36 43691 90
INLIF_1 95 65536 60
INLIF_2 15 65536 60
ISEM_1 40 98304 30
ISEM_2 15 98304 30
KAPS 704 1531904 0
LEM 18 3014656 30
OAM_MEP_DB 2 24576 1
TCAM_IPMF1_80 1404 0 1
TCAM_IPMF3_80 12 0 1
TCAM_PMF_ALL_INTRN_801416 102400 1
TCAM_TNL_TERM 48 16000 178

nfevrier@rtme-acx-48l-03:pfe>
```

For your reference, the following chart describes the difference resources:

Resource	Description
COS_HR_ELEM	COS High Resolution Scheduling Elements
COS_VOQ	Virtual Output Queues
COS_VOQ_CNCTR	Counter Pairs associated with each VOQ
ECMP	Load Balancing Index
EEDB_1	Encapsulation DB for RIF
EEDB_2	Encapsulation DB for Native ARP, SRH base
EEDB_3	Encapsulation DB for Native AC, MPLS Port, SRv6 SID, BFD IPv4 OAM Endpoint
EEDB_4	Encapsulation DB for MPLS tunnel, SIP-tunnel, SRv6 SID
EEDB_5	Encapsulation DB for MPLS tunnel, SRV6 SID
EEDB_6	Encapsulation DB for MPLS tunnel, SRV6 SID
EEDB_7	Encapsulation DB for ARP, Recycle
EEDB_8	Encapsulation DB for AC
EGR_MC_GRP	Multicast group table across all services
EGR_VSI	Bridge Domain / L3 interfaces
ENCAP_EXT	Encapsulation extension associated with encapsulation of outlif(IFL) multicast members
ESEM	VLAN Translation
FAILOVER	Failover ID table associated with protection scenarios
FEC_1	Next-Hop Hierarchy
FEC_2	Next-Hop Hierarchy
FEC_3	Next-Hop Hierarchy
GLEM	Associated with global Outlif (IFL) management
IFL_STATS	IFL statistics, proportional to number of IFL
ING_MC_GRP	Ingress Mcast Groups
ING_VSI	Family IFF and OR Bridge domain table
INLIF_1	Input Logical Interfaces
INLIF_2	Input Logical Interfaces
ISEM_1	In-AC Classification
ISEM_2	Tunnel Termination (MPLS, IP)
KAPS	LPM: v4/v6 routes (ucast/mcast)
LEM	MAC addresses, MPLS labels

Resource	Description
LEXEM_PMF	Large Exact match tables used by IRB Bridge
MYMAC_TCAM	Self Mac entries
OAM_MEP_DB	Maintenance end point database
OAM_NON_ACC_DB	OAM filter is installed to trap OAM packets to inline or control plane
TCAM_EPMF_80	80-bit Egress TCAM entries consumed by egress firewall and junos internal implicit control filter installation (like EVPN MH, MC-LAG, RFC2544 etc.)
TCAM_EXTRN_80	80-bit External TCAM (OP2 based platforms) entries consumed by Ingress firewall configuration
TCAM_IPMF1_80	80-bit ingress firewall entries (stage-1), and implicit control filters
TCAM_IPMF2_80	80-bit ingress firewall entries (stage-2), and implicit control filters
TCAM_IPMF3_80	80-bit ingress implicit control filters (used in stage-3). Examples uRFP counters, sflow, VXLAN MH etc
TCAM_PMF_ALL_EXTRN_80	Aggregate of all External 80-bit tcam entries
TCAM_PMF_ALL_INTRN_80	Aggregate of all internal 80-bit tcam entries
TCAM_TNL_TERM	VTT1 TCAM entries, used for tunnel terminations (proportional to the number of tunnels, IPv6, FTI, etc)
TRUNK_ID	Aggregation group id table
VRRP_SEXEM_1	Small exact match table for VRRP group ID configuration

1.7 In Conclusion

The Junos 23.2 and 23.3 releases introduced important modification in the hardware profiles and the route management. It’s now more flexible and simple from an operation perspective. Hope it helps selecting the best profile for your use-case.

1.8 Useful links

- https://techpost.hpe.com/articles/ptx-fib-compression/article
- https://www.juniper.net/documentation/us/en/software/junos/routing-policy/topics/ref/statement/system-packet-forwarding-options-hw-db-profile.html

1.9 Glossary

- BGP: Border Gateway Protocol
- DRAM: Dynamic Random Access Memory
- cFIB: FIB Compression
- FIB: Forwarding Information Base
- GRT: Global Routing Table
- IFD: Physical Interface
- IFL: Logical Interface
- PFE: Packet Forwarding Engine
- KAPS: KBP Assisted Prefix Search
- KBP: Knowledge Based Processor
- LEM: Large Exact Match
- LPM: Longest Prefix Match
- MDB: Modula DataBase
- QoS: Quality of Service
- RCY: Recycling (interface)
- SRAM: Static Random Access Memory
- TCAM: Ternary Content Addressable Memory
- VLAN: Virtual Local Area Network
- VRF: Virtual Routing and Forwarding instance

1.10 Acknowledgments

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